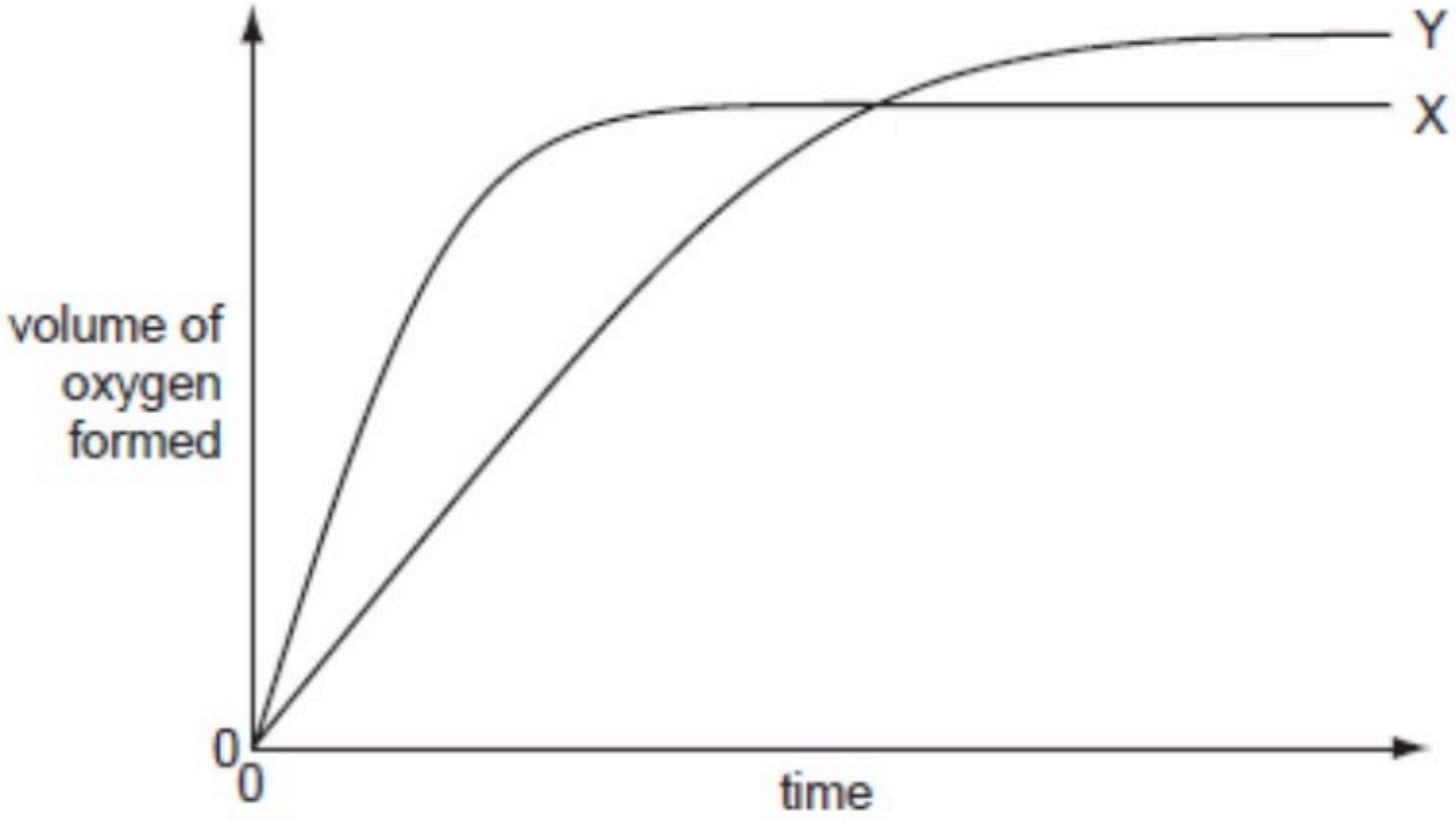
# Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

**Q.1**

In the diagram, curve $X$ was obtained by observing the decomposition of $100\ \text{cm}^3$ of $1.0\ \text{moldm}^{-3}$ hydrogen peroxide, catalysed by manganese (IV) oxide.
Which alteration to the original experimental conditions would produce curve $Y$ ?




**(A)** Adding water

**(B)** Adding some $0.1\ \text{moldm}^{-3}$hydrogen peroxide

**(C)** Using less manganese (IV) oxide

**(D)** Lowering the temperature

**Q.2**

A first order reaction is $50\%$ completed in 20 minutes at $27^\circ\text{C}$ and in $5\ \text{min}$ at $47^\circ\text{C}$. The energy of activation of the reaction is

**(A)** $43.85\ \text{kJ/mol}$

**(B)** $55.14\ \text{kJ/mol}$

**(C)** $11.97\ \text{kJ/mol}$

**(D)** $6.65\ \text{kJ/mol}$

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**Q.3**

A reaction $2A + B \xrightarrow{k} C + D$ is first order with respect to $A$ and 2nd order with respect to $B$. Initial conc. $(t = 0)$ of $A$ is $C_0$ while $B$ is $2C_0$. If at $t = 30$ minutes the conc. of $\mathrm{C_{is}}$ $\mathrm{C_0/4}$ then rate expression at $\mathrm{t = 30}$ minutes is :

**(A)** $R = \frac{7\mathrm{C_0^3 k}}{16}$

**(B)** $R = \frac{27C_0^3\mathrm{k}}{32}$

**(C)** $R = \frac{247C_0^3\mathrm{k}}{64}$

**(D)** $\mathrm{R} = \frac{49\mathrm{kC_0^3}}{32}$

**Q.4**

$$\underset{\text{Sucrose}}{\mathrm{C_{12}H_{22}O_{11}}} + \mathrm{H_2O} \longrightarrow \underset{\text{Glu cose}}{\mathrm{C_6H_{12}O_6}} + \underset{\text{Fructose}}{\mathrm{C_6H_{12}O_6}}$$

Hydrolys is of sucrose is proceeded at a certain temperature and, volume of solution is maintained as 1L. At the beginning, the initial rotation is found $34^\circ$, after $30\ \mathrm{min}$ the total rotation of the solution is 19 and after sufficient time, the total rotation is $-11^\circ\mathrm{C}$. After what time the solution becomes optically inactive ?

**(A)** $135\ \mathrm{min}$

**(B)** $103.7\ \mathrm{min}.$

**(C)** $38.7\ \mathrm{min}$

**(D)** $45\ \mathrm{min}$

**Q.5**

For the two gaeous reactions, following data are given

$$\mathrm{A \longrightarrow B; k_1 = 10^{10} e^{-20000/T}}$$

$$\mathrm{C \longrightarrow D; k_2 = 10^{12} e^{-24606/\ T}}$$

The temperature at which $\mathrm{k_1}$ becomes equal to $\mathrm{k_2}$ is -

**(A)** $400\ \mathrm{K}$

**(B)** $1000\ \mathrm{K}$

**(C)** $800\ \mathrm{K}$

**(D)** $1500\ \mathrm{K}$

**Q.6**

From the following data : $\mathrm{H_2 + I_2 \longrightarrow 2HI}$

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| T(K) | $1/\mathrm{T}\left(\mathrm{K}^{-1}\right)$ | log k |
|---|---|---|
| 769 | $1.3 \times 10^{-3}$ | 2.9 |
| 667 | $1.5 \times 10^{-3}$ | 1.1 |

Suggest which of the following is the correct activation energy for the reaction?

**(A)** $10\ \mathrm{Kcal/mol}$

**(B)** $20\mathrm{cal/mol}$

**(C)** $40\ \mathrm{Kcal/mol}$

**(D)** $80\mathrm{cal/mol}$

**Q.7**

The ionization constant of $\mathrm{NH_4}^+$ in water is $5.6 \times 10^{-10}$ at $25^{\circ}\mathrm{C}$. The rate constant for the reaction of $\mathrm{NH_4^+}$ and $\mathrm{OH^-}$ to form $\mathrm{NH_3}$ and $\mathrm{H_2O}$ at $25^{\circ}\mathrm{C}$ is $3.4 \times 10^{10}\ \mathrm{L\ mol^{-1}\ S^{-1}}$. Which of the following is the correct rate constant for proton transfer from water to $\mathrm{NH_3}$ ?

**(A)** $6.07 \times 10^5\ \mathrm{s^{-1}}$

**(B)** $6.07 \times 10^7\ \mathrm{s^{-1}}$

**(C)** $3.01 \times 10^4\ \mathrm{s^{-1}}$

**(D)** $3.01 \times 10^6\ \mathrm{s^{-1}}$

**Q.8**

The reaction $2A(\mathrm{\ g}) + B_2(\mathrm{\ g}) \rightarrow 2AB(\mathrm{\ g})$ is second order in $\mathrm{A}$ and first order in $B_2$. When 10 mole of $\mathrm{A}$ and $4\ \mathrm{mol} B_2$ are taken in vessel of volume $2dm^3$ together, and the initial rate is $2 \times 10^{-3}$ mole $dm^{-3}s^{-1}$. What will be the rate when half quantity of the $B_2$ has reacted?

**(A)** $4.3 \times 10^{-3}\ \mathrm{mol\ dm^{-3}\ s^{-1}}$

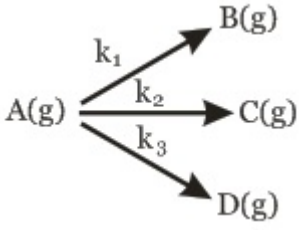
**(B)** $3.6 \times 10^{-4}\ \mathrm{mol\ dm^{-3}\ s^{-1}}$

**(C)** $3.6 \times 10^{-3}\ \mathrm{mol\ dm^{-3}\ s^{-1}}$

**(D)** $3.2 \times 10^{-3}\ \mathrm{mol\ dm^{-3}\ s^{-1}}$

**Q.9**

A gaseous compound, A reacts by three independent first order processes (as shown in figure) with rate constants are 2 x $10^{-3}$, 3 x $10^{-3}$ and 1.93 x $10^{-3}$ $\mathrm{sec^{-1}}$ for products B, C and D respectively. If initially, pure A was taken in a closed container with P = 8 atm, then the partial pressure of B (in atm) after 100 sec from start of experiment.




**(A)** 0.288

**(B)** 0.577

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**(C)** 1.154

**(D)** None of these

**Q.10**

The gas phase decomposition of dimethyl ether follows first order kinetics, $CH_3OCH_3 \rightarrow CH_4 + H_2 + CO_2$. The reaction is carried out in a constant volume container at $500^\circ C$ and has a half-life of $14.5$ minute. Initially only dimethyl ether is present at a pressure of $0.40$ atmosphere. What is the total pressure (approximately) of the system after 12 minute? Assume ideal gas behavior.(anti log of $0.249 = 1.778$ )

**(A)** $0.75$ atm

**(B)** $0.45$ atm

**(C)** $0.35$ cm of Hg

**(D)** $57$ cm of Hg

**Q.11**

Which of the following is/are incorrect regarding activation energy ?

**(A)** Larger the activation energy, smaller is the value of rate constant

**(B)** The higher the $E_a$, higher the value of temperature coefficient, $\frac{k_T+10}{k_T}$.

**(C)** At lower temperature, increase in temperature causes more change in the value of $k$ than at higher temperature.

**(D)** A plot made between $ln\ k$ and $1/T$ gives a straight line of slope $-E_a/R$.

**Q.12**

**Paragraph:**

$A + B \longrightarrow C$

The following data were obtained at $30^\circ C$.

| Experiment | $[A]\ (mol^{-1})$ | Rate |
|---|---|---|
| I | 0.170 | 0.05 |
| II | 0.340 | 0.10 |
| III | 0.680 | 0.20 |

The rate constant for the forward reaction is -

**Question:**

The rate constant

**(A)** $0.294\ h^{-1}$

**(B)** $0.588\ h^{-1}$

**(C)** $0.123\ h^{-1}$

**(D)** $0.210\ h^{-1}$

**Q.13**

**Paragraph:**

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$A + B \longrightarrow C$

The following data were obtained at $30^{\circ}C$.

| Experiment | $[A]\ (mol^{-1})$ | Rate |
|---|---|---|
| I | 0.170 | 0.05 |
| II | 0.340 | 0.10 |
| III | 0.680 | 0.20 |

The rate constant for the forward reaction is -

**Question:**

The equilibrium constant for the reaction is $0.50$. Assuming that the reaction proceeds by one step mechanism. Find the rate constant for the reverse reaction -

**(A)** $0.294\ h^{-1}$

**(B)** $0.588\ h^{-1}$

**(C)** $0.123\ h^{-1}$

**(D)** $0.210\ h^{-1}$

**Q.14**

**Paragraph:**

$A + B \longrightarrow C$

The following data were obtained at $30^{\circ}C$.

| Experiment | $[A]\ (mol^{-1})$ | Rate |
|---|---|---|
| I | 0.170 | 0.05 |
| II | 0.340 | 0.10 |
| III | 0.680 | 0.20 |

The rate constant for the forward reaction is -

**Question:**

In an experiment I, what time reactant A will take to be reduced into $0.0425M$ ?

**(A)** $0.24\ h$

**(B)** $0.48\ h$

**(C)** $0.31\ h$

**(D)** $0.67\ h$

**Q.15**

**Paragraph:**

The rate law expresses the relationship of the rate of a reaction to the rate constant and the concentrations of the reactants raised to some powers for the general reaction $aA + bB \rightarrow cC + dD$ the rate law takes the form $r = k[A]^x[B]y$

where $x$ and $y$ are numbers that must be determined experimentally, $k$ is rate constant and $[A]$ and [B] are concentration of $A$ and $B$ respectively.

**Question:**

Gaseous reaction $A \rightarrow B + C$ follows first order kinetics. Concentration of A changes from $1M$ to $0.25M$ in $138.6$ minutes. Find the rate of reaction when concentration of $A$ is $0.1M$.

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**(A)** $10^{-2}\mathrm{Mmin}^{-1}$

**(B)** $10^{-3}\mathrm{Mmin}^{-1}$

**(C)** $10^{-4}\mathrm{Mmin}^{-1}$

**(D)** None of these

**Q.16**

**Paragraph:**

The rate law expresses the relationship of the rate of a reaction to the rate constant and the concentrations of the reactants raised to some powers for the general reaction $\mathrm{aA}+\mathrm{bB} \rightarrow \mathrm{cC}+\mathrm{dD}$ the rate law takes the form $\mathrm{r}=\mathrm{k}[\mathrm{A}]^{\mathrm{x}}[\mathrm{B}]\mathrm{y}$

where $x$ and $y$ are numbers that must be determined experimentally, $k$ is rate constant and $[A]$ and [B] are concentration of $\mathrm{A}$ and $\mathrm{B}$ respectively.

**Question:**

The initial rate of zero order reaction of the gaseous reaction $\mathrm{A}(\mathrm{g}) \rightarrow 2\ \mathrm{B}(\ \mathrm{g})$ is $10^{-2}\mathrm{Mmin}^{-1}$. If the initial concentration of $\mathrm{A}$ is $0.1\mathrm{M}$, what would be the concentration of $\mathrm{B}$ after $60\ \mathrm{sec}$. ?

**(A)** $0.09\mathrm{M}$

**(B)** $0.01\mathrm{M}$

**(C)** $0.02\mathrm{M}$

**(D)** None of these

**Q.17**

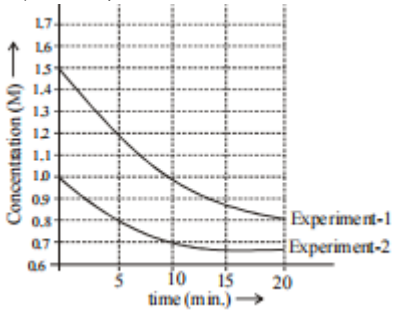
**Paragraph:**

The rate law expresses the relationship of the rate of a reaction to the rate constant and the concentrations of the reactants raised to some powers for the general reaction $\mathrm{aA}+\mathrm{bB} \rightarrow \mathrm{cC}+\mathrm{dD}$ the rate law takes the form $\mathrm{r}=\mathrm{k}[\mathrm{A}]^{\mathrm{x}}[\mathrm{B}]\mathrm{y}$

where $x$ and $y$ are numbers that must be determined experimentally, $k$ is rate constant and $[A]$ and [B] are concentration of $\mathrm{A}$ and $\mathrm{B}$ respectively.

**Question:**

The variation of concentration of A with time in two experiments starting with two different initial concentration of $\mathrm{A}$ is given in the following graph. The reaction is represented as $\mathrm{A}(\mathrm{aq}) \rightarrow \mathrm{B}(\mathrm{aq})$. What is the rate of reaction $(\mathrm{M}/\mathrm{min})$ when concentration of $\mathrm{A}$ in aqueous solution was $1.8\mathrm{M}$ ?




**(A)** $0.072\mathrm{Mmin}^{-1}$

**(B)** $0.036\mathrm{Mmin}^{-1}$

**(C)** $0.1296\mathrm{Mmin}^{-1}$

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**(D)** $1 \mathrm{Mmin}^{-1}$

**Q.18**

Match the column

| Column I | Column II |
|---|---|
| (A) $CH_4 + Cl^* \longrightarrow CH_3^* + HCl$ | (p) Rate of reaction can be predicted with law of mass action expression. |
| (B) $R-X + OH^-(conc.) \longrightarrow R-OH + X^-$ | (q) Rate can be known only by rate law expression. |
| (C) $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ | (r) Molecularity and order of reaction is same. |
| (D) $N_2 + 3H_2 \xrightarrow{\text{Excess Mo}} 2NH_3$ | (s) Rate of reaction remains same with passage of time. |

**(A)** A-p,r; B-p,r; C-q,r; D-q,s;

**(B)** A-p,q; B-r ; C- p,q,s ; D-p,q,s;

**(C)** A-p,q,s; B-p,r ; C- p,q,r,s ; D-p,q,s;

**(D)** A-r,s; B-p,r ; C- p ; D-p,q,s;

**Q.19**

Lead is the final produced formed by a series of changes in which the rate determining stage is the radioactive decay of uranium-238. This radioactive decay is a first order reaction with a half-life of $4.5 \times 10^9$ years. The age of a rock sample is $9.0 \times 10^{A}$ years, (Originally lead free, the molar proportion of uranium of lead is now $1:3$ ). Find the value of $A$.

**Q.20**

The rate of a reaction depends on concentration according to the reaction as follows:

$$-\frac{dC}{dt} = \frac{k_1 C}{1 + k_2 C}$$

The order of the reaction when concentration is very high, is -

## Answers & Solutions

**Q.1 Answer:**

Adding some $0.1\ \mathrm{moldm}^{-3}$hydrogen peroxide

**Solution:**

**Addition of $0.1\ \mathrm{moldm}^{-3} H_2O_2$ dilutes the $H_2O_2$ solution and increases the total amount of $H_2O_2$ present. Hence, the rate of reaction becomes slower (more gentle slope) and the total volume of $O_2$ evolved is higher.**

**Q.2 Answer:**

$55.14\ \mathrm{kJ/mol}$

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**Solution:**

$t_{1/2} = 20$ min at 300 K

$t_{1/2} = 5$ min at 320 K

$$t_{1/2} = \frac{0.693}{A \cdot e^{\frac{-Ea}{300R}}} = 20$$

$$t'_{1/2} = \frac{0.693}{A \cdot e^{\frac{-Ea}{320R}}} = 5$$

$$\frac{t_{1/2}}{t_{1/2}} = 4 = \frac{e^{\frac{-Ea}{320R}}}{e^{\frac{-Ea}{300R}}}$$

$$4 = e^{\frac{Ea}{R}\left(\frac{1}{300} - \frac{1}{320}\right)}$$

$$\ln 4 = \frac{Ea}{R}\left(\frac{20}{300 \times 320}\right)$$

On solving, Ea $= 55303.12$ J

$= 55.3$KJ

**Q.3 Answer:**

$R = \frac{49kC_0^3}{32}$

**Solution:**

| | 2A | + B | $\xrightarrow{k}$ | C | + D |
|---|---|---|---|---|---|
| t = 0 | $C_0$ | $2C_0$ | | 0 | 0 |
| t = 30 | $C_0 - 2x$ | $2C_0 - x$ | | x | x |

**given,** $x = \frac{C_0}{4}$

**After** 30 min, $[A] = C_0 - 2 \times \frac{C_0}{4} = \frac{C_0}{2}$

$$[B] = 2C_0 - \frac{C_0}{4} = \frac{7C_0}{4}$$

$$r = K\left[\frac{C_0}{2}\right]\left[\frac{7C_0}{4}\right]$$

$$\Rightarrow R = \frac{49kC_0^3}{32}$$

**Q.4 Answer:**

103.7 min.

**Solution:**

$k = \frac{2.303}{t} \log \frac{(r_\infty - r_0)}{(r_\infty - r_t)} = \frac{2.303}{30} \log\left(\frac{-11-34}{-11-19}\right) = 1.35 \times 10^{-2}\ \text{min}^{-1}$

**When solution is optically inactive** $r_t = 0$.

**Putting the value of** $r_t$ **in the expression of** $k$.

$t = 103.7$ min.

**Q.5 Answer:**

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$1000\ \mathrm{K}$

**Solution:**

$\mathrm{k_1} = 10^{10}\mathrm{e}^{-20000/\mathrm{T}}$

$$\mathrm{k_2} = 10^{12}\mathrm{e}^{-24606/\mathrm{T}}$$
$$\mathrm{k_1} = \mathrm{k_2}$$
$$10^{10}\mathrm{e}^{-20000/\mathrm{T}} = 10^{12}\mathrm{e}^{-24606/\mathrm{T}}$$
$$\mathrm{e}^{(-20000/\mathrm{T})+(-24606/\mathrm{T})=10^2}$$
$$\mathrm{e}4606/\mathrm{T} = 10^2$$

**On taking log both sides**

$$\frac{4.606}{2.303\ \mathrm{T}} = \log 10^2$$
$$\mathrm{T} = \frac{4606}{4.606} = 1000\ \mathrm{K}$$

**Q.6 Answer:**

$40\ \mathrm{Kcal/mol}$

**Solution:**

$\textbf{Slope} = \frac{(2.9-1.1)}{(1.3-1.5)\times 10^{-3}\ \mathrm{K^{-1}}} = -9 \times 10^3\ \mathrm{K}$

$\mathrm{E_a} = -2.303\mathrm{R}(\text{ slope }) = -2.303 \times 1.99 \times -9 \times 10^3 = 40\ \mathrm{Kcal/mol}$

**Q.7 Answer:**

$6.07 \times 10^5\ \mathrm{s^{-1}}$

**Solution:**

**Proton transfer occurs from water to $\mathrm{NH_3}$.**

$$\mathrm{NH_3 + H_2O \longrightarrow NH_4^+ + OH^-}$$

**Given: Ionization :** $\mathrm{NH_4^+(aq) \rightleftharpoons NH_3 + H^+}, \mathrm{k_f} = 5.6 \times 10^{-10}$

$$\mathrm{k_f} = \frac{[\mathrm{NH_3}]\,[\mathrm{H^+}]}{\left[\mathrm{NH_4^+}\right]}$$

**Neutralisation:** $\mathrm{NH_4^+ + OH^- \rightleftharpoons NH_3 + H_2O}$

**Rate constant** $= 3.4 \times 10^{10}\ \mathrm{L\ mol^{-1}\ s^{-1}} = \mathrm{k_f}$

**Equilibrium constant,** $\mathrm{K_{eq}} = \frac{[\mathrm{NH_3}][\mathrm{H_2O}]}{\left[\mathrm{NH_4^+}\right][\mathrm{OH^-}]} = \frac{[\mathrm{NH_3}]}{\left[\mathrm{NH_4^+}\right][\mathrm{OH^-}]} = \frac{\mathrm{k_f}}{\mathrm{k_b}}$

**The reverse of neutralisation** $\mathrm{k_b}$ **(backward)** $= \frac{\left[\mathrm{NH_4^+}\right][\mathrm{OH^-}]}{[\mathrm{NH_3}]} = ?$

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**From $k_f$ expression** $\left[NH_4^+\right] = \frac{[NH_3]\left[H^+\right]}{k_i}$

**Putting value of** $[NH_4{}^+]$ **in** $K_{eq}$ **.**

$$k_{eq} = \frac{k_i}{[H^+][OH^-]} = \frac{5.6\times10^{-10}}{10^{-14}} = 5.6 \times 10^4$$

$$\text{But, } K_{eq} = \frac{k_f}{k_b} \quad \text{or } K_b = \frac{k_f}{k_{eq}} = \frac{3.4\times10^{10}}{5.6\times10^4} = 6.07 \times 10^5 \text{ s}^{-1}$$

**Q.8 Answer:**

$3.6 \times 10^{-4}$ mol $dm^{-3}$ $s^{-1}$

**Solution:**

$R_f = k\left[A_i\right]^2\left[B_{2i}\right]$

$R_f = k\left[A_f\right]^2\left[B_{2f}\right]$

$R_f = 2 \times 10^{-3}$ **mol** $dm^{-3}$ $s^{-1}$

$[A_f] = \frac{10}{2} = 5$ **mol** $dm^{-3}$

$[B_{2f}] = \frac{4}{2} = 2$ **mol dm**

$R_f = ?$

$[A_f] = \frac{6}{2} = 3$ **mole** $dm^{-3}$

$[B_{2f}] = \frac{2}{2} = 1$ **mole** $dm^{-3}$

$R_f = \frac{2\times10^{-3}}{5^2\times2} \times 3^2 \times 1$

$= 3.6 \times 10^{-4}$ **mole** $dm^{-3}$ $s^{-1}$

**Q.9 Answer:**

1.154

**Solution:**

$$A(g)\begin{cases}\xrightarrow{k_1} B(g)\\ \xrightarrow{k_2} C(g)\\ \xrightarrow{k_3} D(g)\end{cases}$$

$K_{net} = k_1 + k_2 + k_3 = 6.93 \times 10^{-3}\text{sec}^{-1}$

$t_{1/2} = \frac{0.693}{6.93\times10^{-3}} = \frac{693}{6.93} = 100$ sec.

**i.e., After 100 sec. A(g) will decompose half its initial moles**

**i.e., Partial pressures of B, C and D together will be 4 atm**

$P_B = \frac{k_1}{k_1 + k_2 + k_3} \times 4$

$= \frac{2\times10^{-3}}{6.93\times10^{-3}} \times 4 = 1.154$ atm

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**Q.10 Answer:**

$0.75$ atm

**Solution:**

$$\underset{\text{Initial P}}{\underset{0.40 \text{ atm}}{CH_3.OCH_3}} \rightarrow \underset{P}{CH_4} + \underset{P}{H_2} + \underset{P}{CO_2}$$

**For a I order reaction** $K = \frac{2.303}{t} \log_{10} \frac{a}{a-x}$

**For ideal gas behavior Mole** $\alpha$ **Pressure (at constant** $V$ **and T)**

$\therefore a \alpha 0.40; (a-x) \alpha (0.40 - P)$

$$\frac{0.693}{14.5} = \frac{2.303}{12} \log_{10} \frac{0.40}{(0.40 - P)}$$

$$P = 0.175 \text{ atm}$$

**Pressure of ether decomposed** $= 0.175$

**Total pressure** $= 0.40 - P + P + P + P$

$= 0.40 + 2P = 0.40 + 2 \times 0.175$

$= 0.75 \text{ atm} = 57 \text{ cm}$

**Q.11 Answer:**

The higher the $E_a$, higher the value of temperature coefficient, $\frac{k_T+10}{k_T}$.

**Solution:**

$$\frac{k_T + 10\,K}{k_T} = \exp\left(\frac{(10\,K)E_a}{RT(T+10\,K)}\right)$$

A plot between $\ln k$ and $\frac{1}{T}$ gives straight line.

**Q.12 Answer:**

$0.294\ h^{-1}$

**Solution:**

**(i)** $r_f = k_f[A]^n$, **n is the order of reaction.**

| | | | |
|---|---|---|---|
| | 0.170 | 0.05 | |
| twice | 0.340 | 0.10 | twice |
| gain twice | 0.680 | 0.20 | twice |

**It suggests first order reaction. From first** $I: r_f = 0.05, [A] = 0.10M$

| | | | |
|---|---|---|---|
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| again twice | 0.680 | 0.20 | twice |

It suggests first order reaction.

From first $I: r_f = 0.05, [A] = 0.10M$

$r_f = k_f[A] \Rightarrow 0.05 = k_f \times 0.170$

$k_f = 0.294\ h^{-1}$

$K_c = \frac{k_f}{k_b}$ **or** $k_b = \frac{k_f}{K_c} = \frac{0.294}{0.50}$

$k_b = 0.588\ h^{-1}$

**(ii) Expt. I: Initial conc.** $= 0.170M$

**Final conc.** $= 0.0425M$ **(one fourth)**

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**Order of reaction is 1.**

**(iii) Half-life time** $= \frac{0.693}{\mathrm{k}} = \frac{0.693}{0.294} = 0.24$

**One fourth will takes** $2 \times \mathrm{t}_{1/2} = 0.24 \times 2 = 0.48$

**Q.13 Answer:**

$0.588\ \mathrm{h}^{-1}$

**Solution:**

**(i)** $\mathrm{r_f} = \mathrm{k_f}[\mathrm{A}]^{\mathrm{n}}$, **n is the order of reaction.**

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It suggests first order reaction.

From first $\mathrm{I}: \mathrm{r_f} = 0.05, [\mathrm{A}] = 0.10\mathrm{M}$

$r_{\mathrm{f}} = \mathrm{k_f}[\mathrm{A}] \Rightarrow 0.05 = \mathrm{k_f} \times 0.170$

$\mathrm{k_f} = 0.294\ \mathrm{h}^{-1}$

$\mathrm{K_c} = \frac{\mathrm{k_f}}{\mathrm{k_b}}$ **or** $\mathrm{k_b} = \frac{\mathrm{k_f}}{\mathrm{K_c}} = \frac{0.294}{0.50}$

$\mathrm{k_b} = 0.588\ \mathrm{h}^{-1}$

**(ii) Expt. I: Initial conc.** $= 0.170\mathrm{M}$

**Final conc.** $= 0.0425\mathrm{M}$ **(one fourth)**

**Order of reaction is 1.**

**(iii) Half-life time** $= \frac{0.693}{\mathrm{k}} = \frac{0.693}{0.294} = 0.24$

**One fourth will takes** $2 \times \mathrm{t}_{1/2} = 0.24 \times 2 = 0.48$

**Q.14 Answer:**

$0.48\ \mathrm{h}$

**Solution:**

**(i)** $\mathrm{r_f} = \mathrm{k_f}[\mathrm{A}]^{\mathrm{n}}$, **n is the order of reaction.**

| | | | |
|---|---|---|---|
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| twice | 0.340 | 0.10 | twice |
| gain twice | 0.680 | 0.20 | twice |

**It suggests first order reaction. From first** $\mathrm{I}: \mathrm{r_f} = 0.05, [\mathrm{A}] = 0.10\mathrm{M}$

| | | | |
|---|---|---|---|
| | 0.170 | 0.05 | |
| twice | 0.340 | 0.10 | twice |
| again twice | 0.680 | 0.20 | twice |

It suggests first order reaction.

From first $\mathrm{I}: \mathrm{r_f} = 0.05, [\mathrm{A}] = 0.10\mathrm{M}$

$r_{\mathrm{f}} = \mathrm{k_f}[\mathrm{A}] \Rightarrow 0.05 = \mathrm{k_f} \times 0.170$

$\mathrm{k_f} = 0.294\ \mathrm{h}^{-1}$

$\mathrm{K_c} = \frac{\mathrm{k_f}}{\mathrm{k_b}}$ **or** $\mathrm{k_b} = \frac{\mathrm{k_f}}{\mathrm{K_c}} = \frac{0.294}{0.50}$

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$k_b = 0.588\ h^{-1}$

**(ii) Expt. I: Initial conc.** $= 0.170M$

**Final conc.** $= 0.0425M$ **(one fourth)**

**Order of reaction is 1.**

**(iii) Half-life time** $= \frac{0.693}{k} = \frac{0.693}{0.294} = 0.24$

**One fourth will takes** $2 \times t_{1/2} = 0.24 \times 2 = 0.48$

**Q.15 Answer:**

$10^{-3}Mmin^{-1}$

**Solution:**

**Half-life of a first-order reaction is constant.**

**In** $138.6$ **min, concentration reduced to** $\frac{1}{4}$ **of the original.**

$$\therefore \text{ Half life of reaction } = \frac{138.6}{2} = 69.3 \text{ min}$$

$$\frac{0.693}{K} = t_{1/2}$$

$$\Rightarrow K = \frac{0.693}{69.3} = 10^{-2} \text{ min}^{-1}$$

$$\text{rate } = K[A]$$

$$= 10^{-2} \times 10^{-1} M \overset{-1}{\text{min}}$$

$$\Rightarrow 10^{-3}Mmin^{-1}$$

**Q.16 Answer:**

$0.02M$

**Solution:**

$r = K$ **and** $[A]_t = [A]_0 - Kt = 0.1 - 10^{-2} \times 1.0 = 0.1 - 0.01 = 0.09M$

**Concentration of B** $= 0.01 \times 2 = 0.02M$

**Q.17 Answer:**

$0.072Mmin^{-1}$

**Solution:**

**Rate law** $r = k[A]^m$

$r_1 = \left(\frac{0.3}{5}\right) M/min$ **from graph ;** $r_2 = \left(\frac{0.2}{5}\right) M/min$ **from graph**

$$\frac{r_1}{r_2} = \frac{3}{2} = \left\{\frac{[A]_1}{[A]_2}\right\}^m = \left(\frac{1.5}{1}\right)^m \Rightarrow m = 1$$

$$\frac{0.3}{5} = k(1.5) \Rightarrow k = \frac{1}{25} \text{ min}^{-1}$$

$$r = k \times [A] \Rightarrow \frac{1}{25} \times 1.8 = 0.072Mmin^{-1}$$

**Q.18 Answer:**

A-p,r; B-p,r; C-q,r; D-q,s;

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**Solution:**

**((A) - p,r, (B) - p,r, (C) - q,r, (D) - q,s)**
**Reactions A and B are elementary reactions. C and D are complex reactions. Ammonia formation is zero order reaction.**

**Q.19 Answer:**

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**Solution:**

**Molar ratio of** $\mathrm{U}:\mathrm{Pb}=1:3=\frac{1}{1+3}:\frac{3}{1+3}=\frac{1}{4}:\frac{3}{4}$
**Therefore, amount of U left is** $\frac{1}{4}$ **of the original**
$\xrightarrow{t_{1/2}} \frac{1}{2} \xrightarrow{t_{1/2}} \frac{1}{4}$
**Hence, the age of the rock sample is** $2 \times t_{1/2}$ **i.e.,** $9.0 \times 10^{9}$ **years**

**Q.20 Answer:**

—

**Solution:**

$-\frac{dC}{dt}=\frac{k_1 C}{1+k_2 C}$ **or** $\frac{k_1 C}{\frac{1}{C}+k_2}$
**If** $C$ **is very high** $\frac{1}{C}$ **can be assumed negligibly small.**
**Thus,** $-\frac{dC}{dt}=\frac{k_1}{k_2}=$ **constant and the order of reaction is zero.**

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